IBM Z DATATHON 2026 · TEAM ARISTOCATS

FelineCast

Predictive urban colony analytics & targeted TNR logistics

Data-driven population intervention on IBM LinuxONE



Track Wildcard: Beyond the Obvious

Platform LinuxONE Community Cloud (s390x)

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THE CALIFORNIA CRISIS

Ecological impact & overpopulation

Why reactive animal control fails free-roaming colonies



Ecological loss

Colonies drive urban biodiversity loss, high outdoor mortality and shelter euthanasia.

500,000+

feral cats in Los Angeles County alone

Wallace & Levy, 2006

<3.5%

of unmanaged community cats are sterilized

Wallace & Levy, 2006

82–91%

of owned pet cats are sterilized; intact strays drive kitten surges

Levy & Crawford, 2004

BASELINE COLONY ESTIMATE

Households × 0.07 × 3.2

≈ Households × 0.22

~7% of households feed an average of 3.2 unowned cats

(Kass et al., 2013)

TIPPING POINT

≥ 70–75%

coverage needed to reverse colony growth

Boone et al., 2019

APPROACH

Data methodology & statistical modeling

Addressing count overdispersion with negative binomial regression



01

Raw shelter data

LBACS records 2017–2026 + U.S. Census households across 11 ZIP codes (90802–90815)

02

Filter pipeline

27,000+ records narrowed to living stray cat intakes (STRAY) and TNR dispatches

03

Dispersion check

Monthly ZIP counts show Var(Y) far above E[Y]: classic overdispersion

04

Negative binomial fit

A dispersion parameter keeps standard errors honest, unlike Poisson

Why not deep learning?

ZIP-level tabular time series lacks the sample depth neural networks need; they would overfit severely.

Why not standard decision trees?

Unconstrained trees produce stepped, discontinuous bounds that can’t extrapolate smooth reproductive growth curves.

PLATFORM

Enterprise system architecture on IBM LinuxONE

High-throughput municipal ingestion on s390x architecture

```
>>> socket.gethostname
'ibmzhackathonserver'
>>> platform.machine
's390x'
```



Live kernel output · LinuxONE Community Cloud (IBM Z)

Prerequisites verified

Completed the mandatory Predictive AI Hybrid Scenarios and Predictive AI + Generative AI labs on LinuxONE Community Cloud.

Mainframe environment

Deployed and trained inside LinuxONE Jupyter Lab ML, running natively on IBM Z s390x hardware.

Enterprise synergies

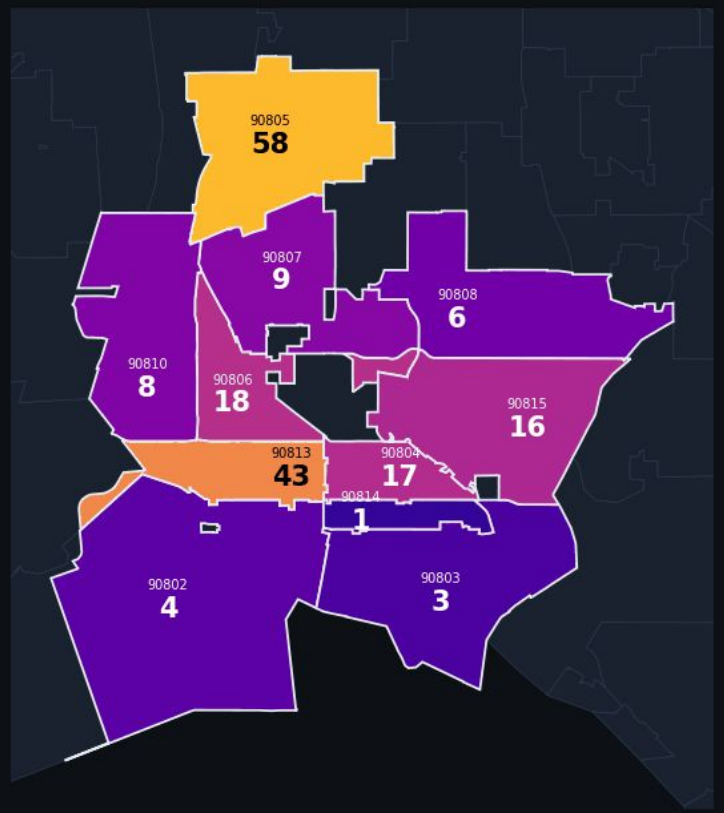
Designed to scale to concurrent 311 calls and shelter telemetry across city departments, with mission-critical reliability and data privacy.

Oct 2025

Stray + TNR cat intakes per month, by ZIP

ACTUAL

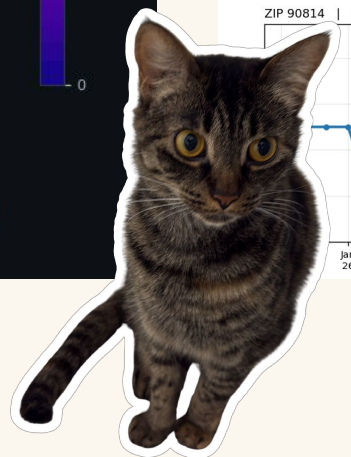
recorded intakes

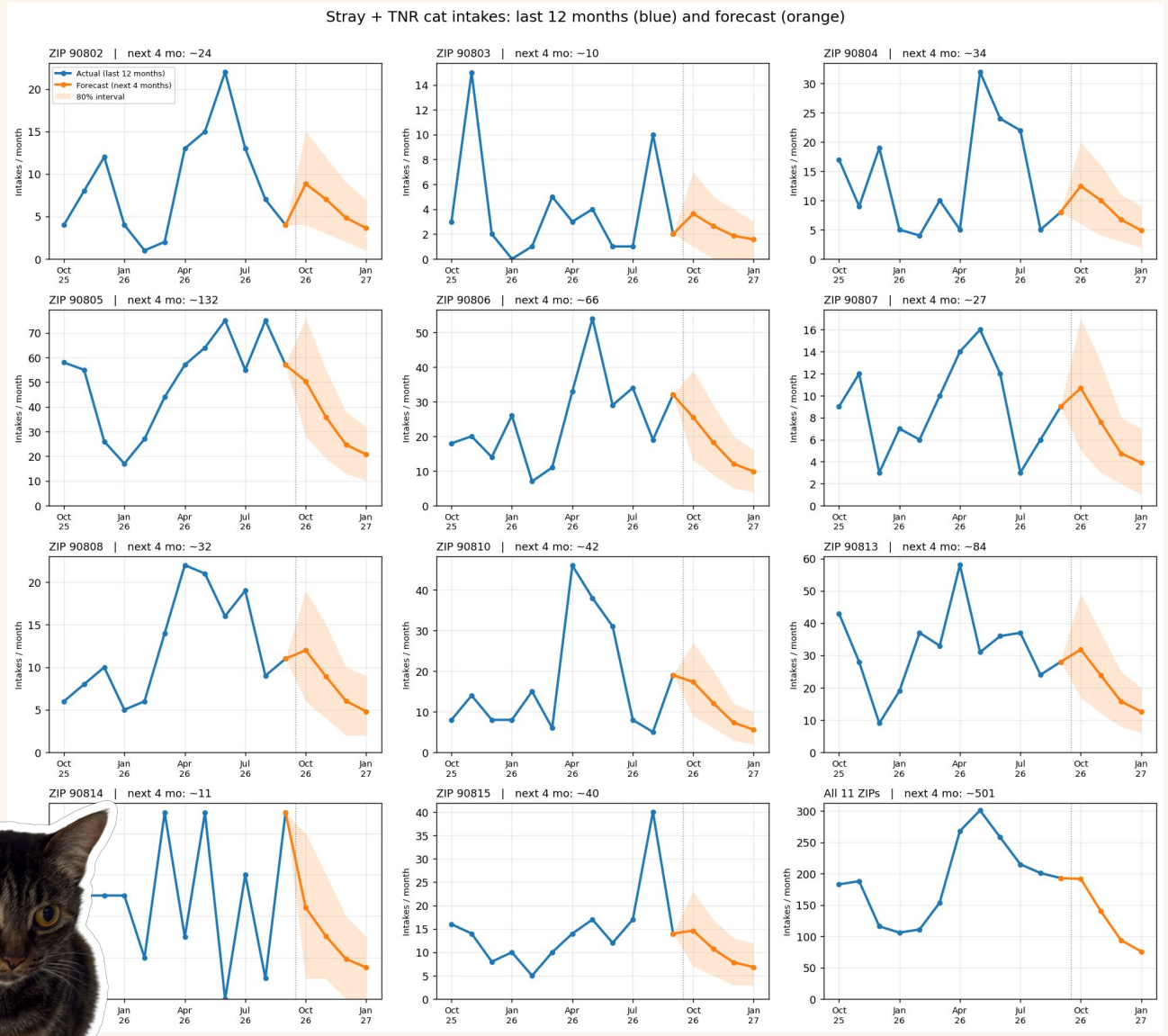




past 12 months

predicted 12 months





RESULTS

Model findings & field dispatch demo

Translating risk scores into actionable TNR logistics

HOTSPOT

ZIP 90805

North Long Beach

~6,300 estimated community cats across 28,213 households, with the strongest seasonal intake velocity.

Dynamic risk matrix

Tracks monthly intake velocity against baseline colony estimates to flag “Red Zones” where sterilization lags breeding surges.



TRAPS TO DEPLOY

= Forecast × 1.5

CLINIC VOUCHERS

= Traps × 0.8

SAMPLE DISPATCH · OCT 2026 FORECAST

ZIP	Forecast	Traps	Vouchers
90805 · top	50.4	76	61
90806	25.5	38	30
90804	12.5	19	15
90802	8.8	13	10
90803	3.6	5	4

IMPACT

Societal impact & nationwide scalability

Transforming reactive sheltering into proactive conservation

Taxpayer efficiency

Shifts city spending from kennels, intake processing and euthanasia to field-level sterilization and stabilization.

Wildlife conservation

Suppresses unmanaged seasonal litters, reducing predation on native wildlife and vulnerable bird populations.

Nationwide scalability

Built on standardized open-data intake fields, ready to replicate in cities like Los Angeles, Austin and Miami.

The hybrid mainframe advantage: scalable, continuous municipal risk forecasting backed by enterprise reliab

